

Fitness & Protein Tracker - Complete Project Architecture

Tech Stack

Frontend: React
Styling: Tailwind CSS
Backend: Node.js + Express.js
Database: PostgreSQL
ORM: Prisma
API Requests: Axios
Authentication: JWT + bcrypt
Charts: Chart.js
File Uploads: Multer + Cloudinary
Deployment: Vercel, Render, Neon PostgreSQL

High-Level Architecture

User → React Frontend → Axios → Express Backend → Services (Authentication, Protein Tracking, Meal Recommendation, Analytics, Nutrition API Integration) → PostgreSQL Database

Core Features

1. Authentication (Signup/Login/Logout)
2. User Fitness Profile
3. Food Database
4. Protein Tracking
5. Nutrition API Integration
6. Protein Goal System
7. Smart Meal Recommendation Engine
8. Daily Meal Generator
9. Weight Tracker
10. Water Tracker
11. Workout Tracker
12. Streak System
13. Dashboard Analytics
14. Progress Photos

Personalized Meal Recommendation

During onboarding ask: Age, Gender, Height, Weight, Goal, Activity Level, Diet Type (Vegetarian, Non-Vegetarian, Vegan, Eggetarian).

Recommendations are generated using: Diet Type + Fitness Goal + Remaining Protein Requirement.

Database Tables

users
user_profiles

foods
food_logs
weight_logs
water_logs
workouts
streaks
progress_photos

Backend Folder Structure

backend/
■ controllers
■ routes
■ middleware
■ services
■ prisma
■ models
■ utils
■ config
■ app.js
■ server.js

Frontend Folder Structure

frontend/
■ src
■ pages
■ components
■ services
■ hooks
■ context
■ layouts
■ App.jsx

Development Roadmap

Phase 1: Authentication, PostgreSQL, User Profiles, Food Logging
Phase 2: Nutrition API Integration, Protein Calculation, Daily Goals
Phase 3: Meal Recommendation Engine, Veg/Non-Veg Logic, Meal Generator
Phase 4: Weight Tracking, Water Tracking, Workout Tracking
Phase 5: Dashboard Charts, Streak System, Deployment
Phase 6: AI Coach, Grocery Planner, Recipe Recommendations